Movie Industry Strategic Analysis

SUBTITLE: IDENTIFYING WHAT TYPES OF FIILMS ARE DOING BEST"

TEAM:

CHARITY NDUATI

CLARE SIMIYU

JAKLINE MWAU

DORCAS NDAMBUKI

VICTOR GITHINJI

DATE: 20/MARCH/2026

Business Context and Problem Statement

- Company is **launching a new movie studio** with **no prior film industry experience**
- Core business goal: **make profitable films** while **minimizing financial risk**
- Key questions the analysis answers:
- Which **genres** generate the highest revenue & ROI?
- What is the **optimal production budget range**?
- When is the **best time of year** to release movies?
- Ultimate objectives:
- Reduce risk of investing in the wrong kinds of films
- Choose the right types of films
- Maximize profit and audience reach

Data Sources and Scope

•We used **three complementary datasets** (2010–2018 focus):

1.Box Office Mojo (bom.movie_gross) → domestic + foreign gross revenue, studio

2.The Numbers (tn.movie_budgets) → production budget, worldwide gross → allows ROI calculation

3.IMDb (im.db) → genres, runtime, average rating, number of votes, release year

•**Most important metrics created**:

•Worldwide Gross

•Profit = Worldwide Gross − Production Budget

•ROI = (Profit / Budget) × 100%

•Median & mean profit / ROI per genre, per budget bucket, per release month

Methodology and Data Preparation

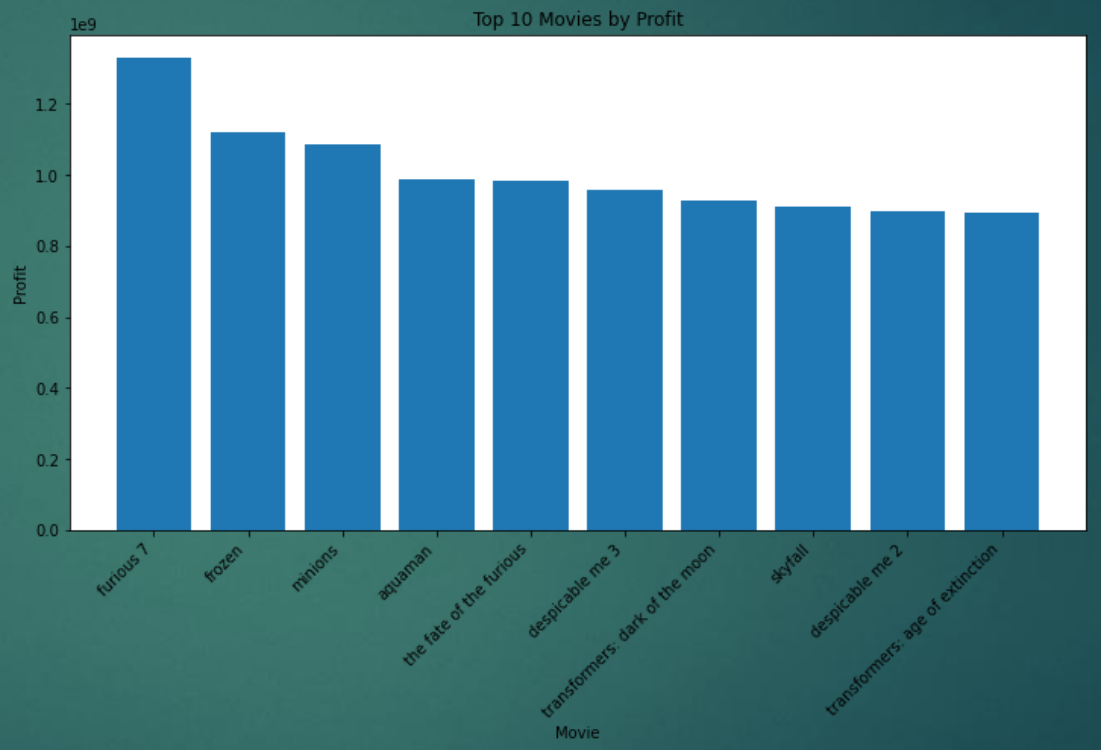
- **Main steps**:
- Loaded & merged datasets (BOM + TN + IMDb)
- Cleaned: standardized currency, handled missing values, parsed dates
- Created derived columns: Profit, ROI, Release Month, Budget buckets
- Aggregated by: Genre (primary genre), Release Year, Release Month, Budget range
- Visualized using bar plots, line plots, box plots, scatter plots

Visualizations

1.Top 10 movies by profit

Insight: Profit Distribution Among Top 10 Movies

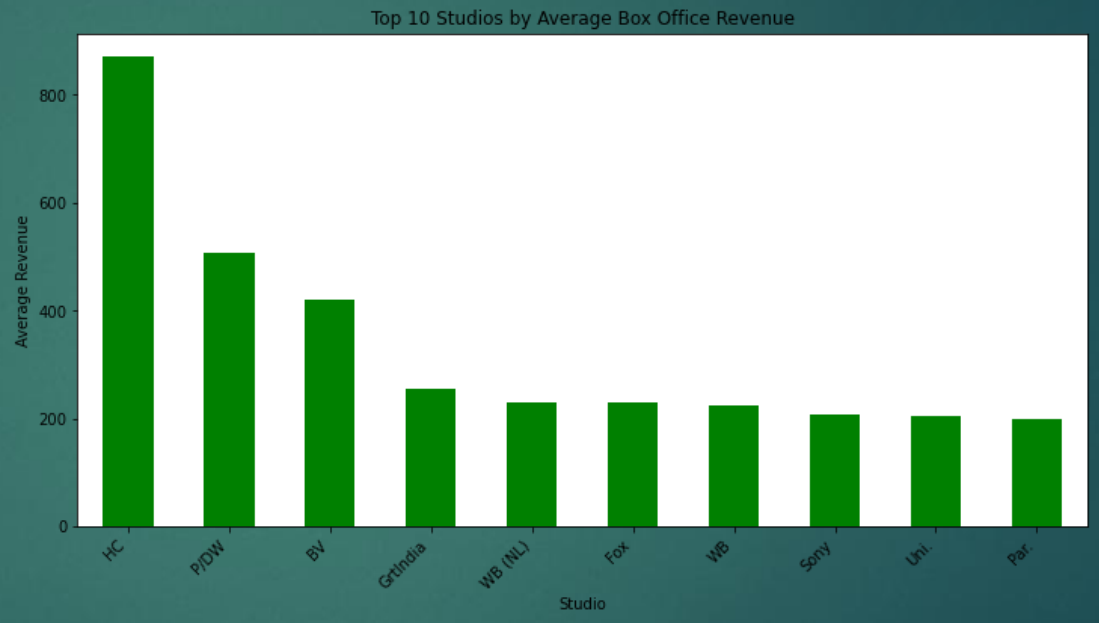
The bar chart reveals that profitability among the top 10 movies is not evenly distributed. A number of movies generate higher profits compared to the rest.



2.Top 10 Studios by average box office revenues

Insight:

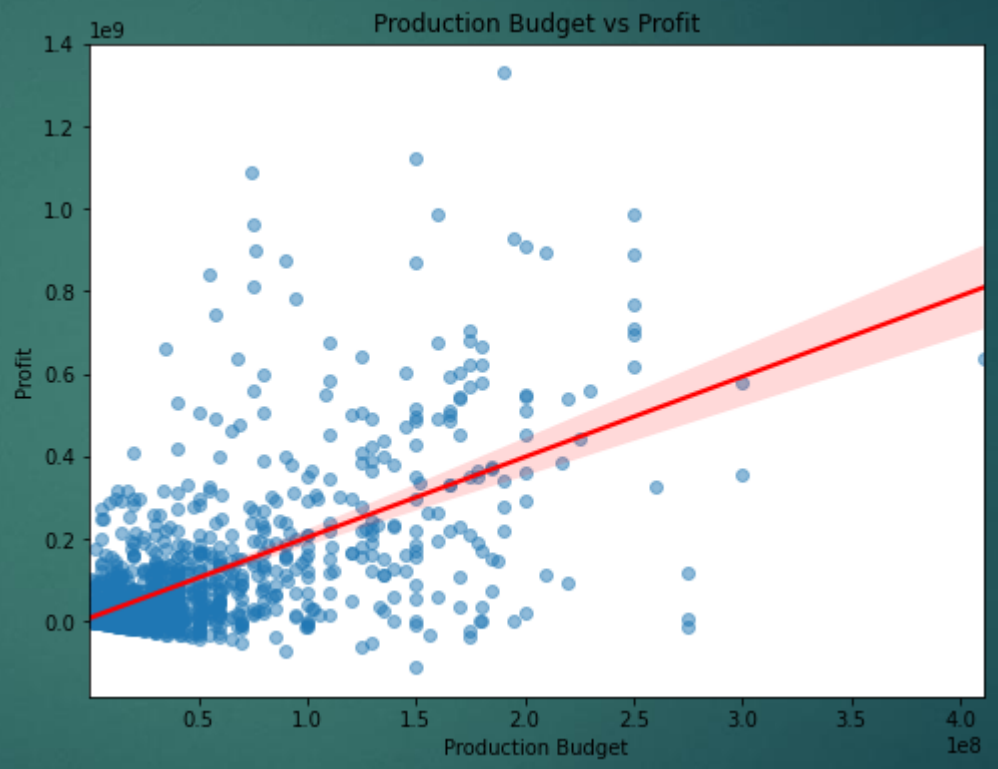
The graph above analyzes the top 10 production studios in gross profit between 2010 to 2018. This shows the best peformers in the film industry, where our company can focus on the type of films they produce.



3. Production budget vs profit

We use a redplot to see the "Line of Best Fit." This helps us understand if spending an extra $10M actually results in more revenue.s.

Insight: There is a "diminishing return" on very high budgets. The most consistent success happens in the $30M–$70M range.



Recommendations

- 1. Invest in High-Impact Productions
- Since profits are concentrated among a few standout films, studios should prioritize projects with strong market potential, such as established franchises, sequels, or films with proven audience appeal.
- 2. Leverage Data-Driven Decision Making
- Use historical profit data, genre trends, audience preferences, and release timing analysis to guide production and marketing strategies.
- 3. Optimize Marketing Strategies
- Given the large profitability gap, aggressive and well-targeted marketing campaigns can help elevate a film's performance into the highest profit tier.
- 4. Diversify Investment Portfolio
- While aiming for blockbuster hits, studios should balance risk by investing in a mix of high-budget and mid-budget films to ensure more stable overall returns.
- 5. Analyze Key Success Drivers
- Further analysis should be conducted on factors such as genre, production budget, release season, and star power to understand what differentiates the highest-profit films from the rest.

Next Steps

- Incorporate post-2019 streaming data, marketing spend, director/star track record

Conclusion

- Analysis of box office data (primarily 2010–2018) reveals that Action, Adventure, Sci-Fi, and Animation genres consistently deliver the highest worldwide revenue and strongest international performance. Family animation and franchise-driven action-adventure films show particularly reliable returns, while mid-budget projects in the $25–60 million range achieve the best risk-adjusted ROI. Extremely high-budget films (> $150 M) produce both massive hits and costly flops, making disciplined budgeting essential.
- Release timing strongly influences success: the May–July summer window and November–December holiday period generate the highest median profits due to school breaks, family audiences, and gift-giving seasons. Off-peak months (January–February, August–September) are best reserved for smaller releases, counter-programming, or streaming-focused titles.
- Strategic recommendation: Build a core slate of 2–3 $40–70 million action-adventure or family animation films per year targeting summer and holiday windows, supplemented by 3–5 lower-budget ($10–40 M) projects for diversification and strong ancillary/streaming revenue. Selective $100 M+ tentpoles should only be pursued with proven franchise value, global star power, and robust international pre-sales. This balanced approach maximizes theatrical upside while controlling financial risk in a competitive market.